



Agnel Charities
Fr. C. Rodrigues Institute of Technology, Vashi
(An Autonomous Institute & Permanently Affiliated to University of Mumbai)
Department of Electronics and Telecommunication Engineering

End sem examination BE Sem VIII Major Project II

**Project Title: Adaptive Learning Multimodal AI Music
Recommendation System**

Group Number: Internal3 (I3)

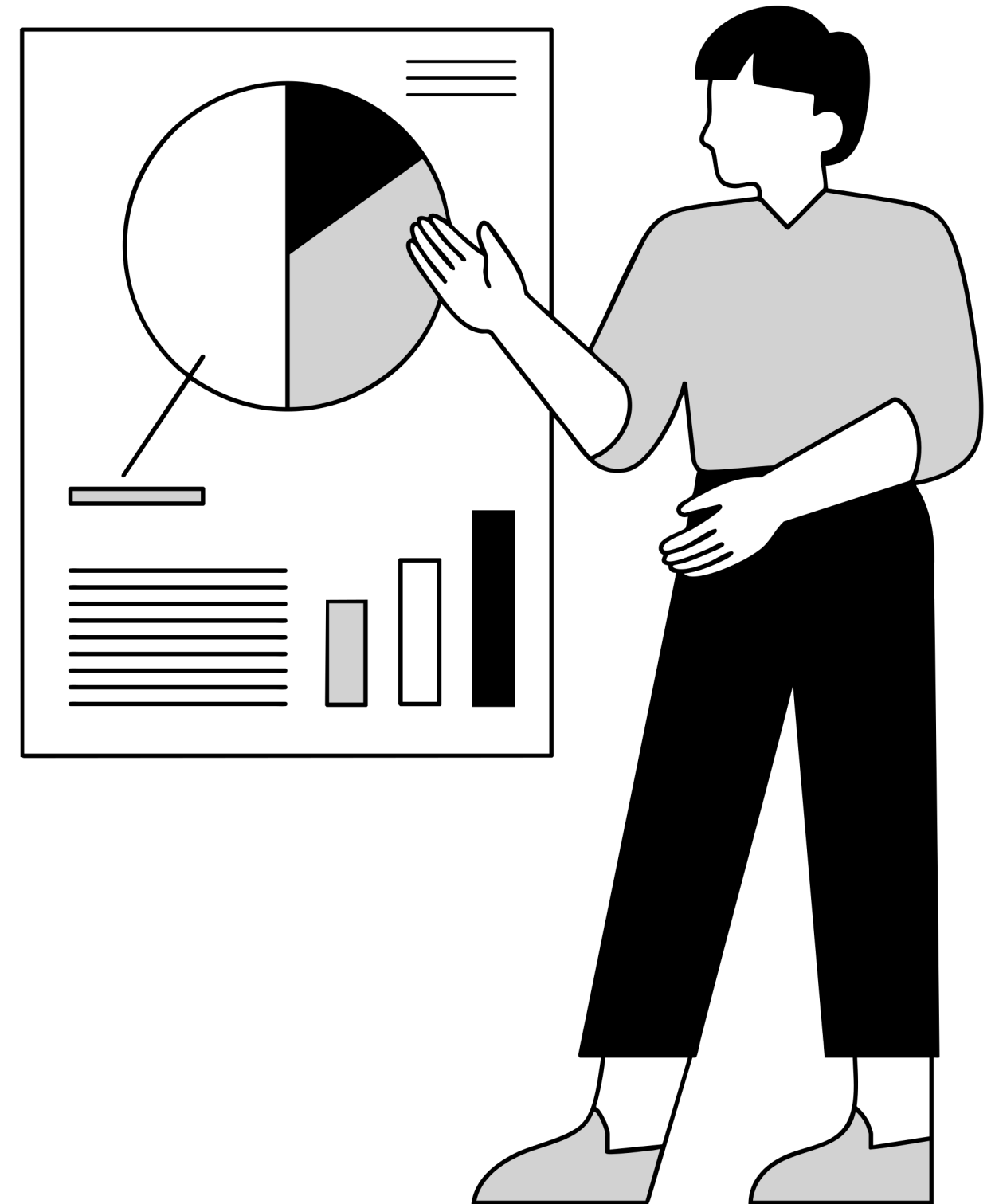
Guided By: Prof Yogesh Chandurkar

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Presentation Outline

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Problem Overview

Music affects how we feel and perform daily, influencing our mood, stress levels, focus, and overall well-being. Therefore, identifying the right music at the right time becomes essential.

Traditional music recommendation systems rely only on:

- Listening history
- Genres & ratings

They ignore emotional and physiological state of the user. Recommendations remain static and context-unaware

Problem Statement

How can we design an intelligent, adaptive music recommendation system that integrates psychological and physiological data to deliver truly personalized, context-aware music suggestions?

Project Objectives

- 1. To capture physiological and psychological data by collecting Heart Rate (HR) and stress inputs to understand users' physical and emotional states.**
- 2. To apply AI-based analysis to process and classify user conditions based on the collected data.**
- 3. To recommend personalized music according to the identified emotional and physiological states to enhance mood and reduce stress.**
- 4. To incorporate adaptive learning by utilizing user feedback and physiological signals to continuously improve recommendation accuracy.**
- 5. To validate system performance through experimental testing with real users to evaluate effectiveness in mood enhancement and stress management.**

Literature Survey.

Title	Authors	Objective	Methodology	Advantages	Year
A Review on the use of Machine Learning Techniques in Music Recommendation System for Healthcare Management	Yogesh P. Pingle, Lakshmappa K. Ragha	Compares supervised & unsupervised ML (SVM, RF, RNN, LSTM, RL, etc.) for emotion- based recommendations.	Taxonomy of systems; highlights models with up to 98% accuracy.	Review ML techniques for healthcare music recommendation.	2023
EMO-Music: Emotion Recognition Based Music Therapy with Deep Learning on Physiological Signals	Hanzhe Guo, Yifei Qi, Jiawen Zhang, et al.	Real-time smartwatch-based emotion detection & music therapy.	Huixin watch data (HR, GSR) processed via BERT time-series model; integrated with mobile/watch app.	70.8% accuracy; real-time emotion-adaptive music.	2024

Literature Survey.

Title	Authors	Objective	Methodology	Advantages	Year
A Personalized Music Recommendation System	Shravanthi K, Sneha Rao S, Swati Joshi, Vandana R; Prof. Maya B. S.	Recommend songs using genre classification & collaborative filtering.	CNN for genre detection; popularity & item-based collaborative filtering.	Combines multiple recommendation types; handles large datasets.	2019
Musical Therapy for Stress Reduction Using Machine Learning and IoT	Achal Mate, Pushkar Narkhede, Pooja Kulkarni, Harshad Lokhande, Jayashree Prasad, Rajesh Prasad	Create an IoT- based wearable + ML system to detect stress in real time and recommend Indian ragas for relaxation.	Sensors track heart rate, BP, motion; ML model predicts stress; Android app suggests suitable ragas.	Real-time personalized therapy; blends AI with traditional music; useful in healthcare and workplaces.	2019

Sustainable Development Goal (SDG)



- **SDG MAPPING – SDG 3 (GOOD HEALTH AND WELL-BEING)**

The Music recommendation system contributes to **SDG 3** by integrating multimodal affective computing for real-time mental health support.

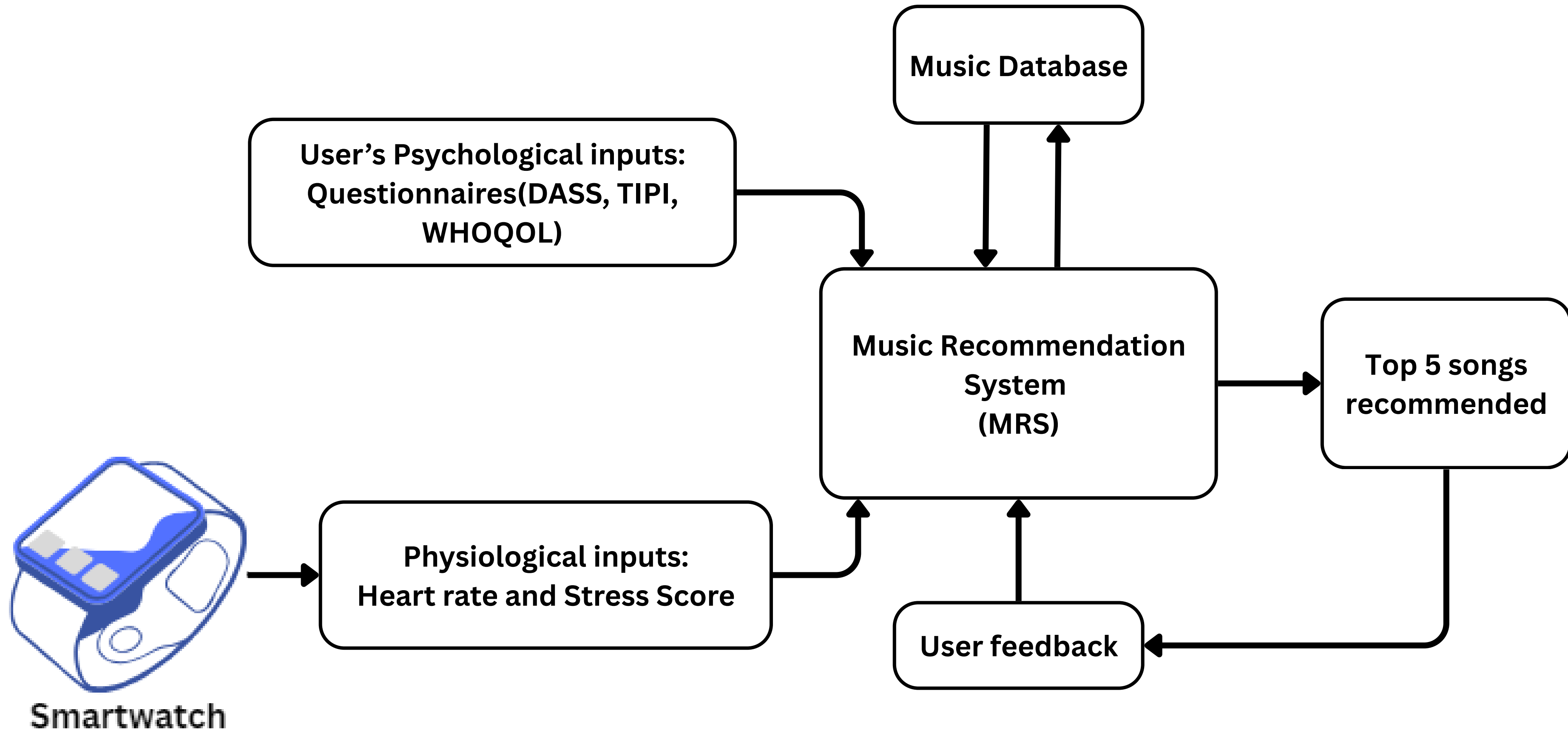
Focuses on:

- Mental health improvement
- Stress reduction
- Emotional well-being

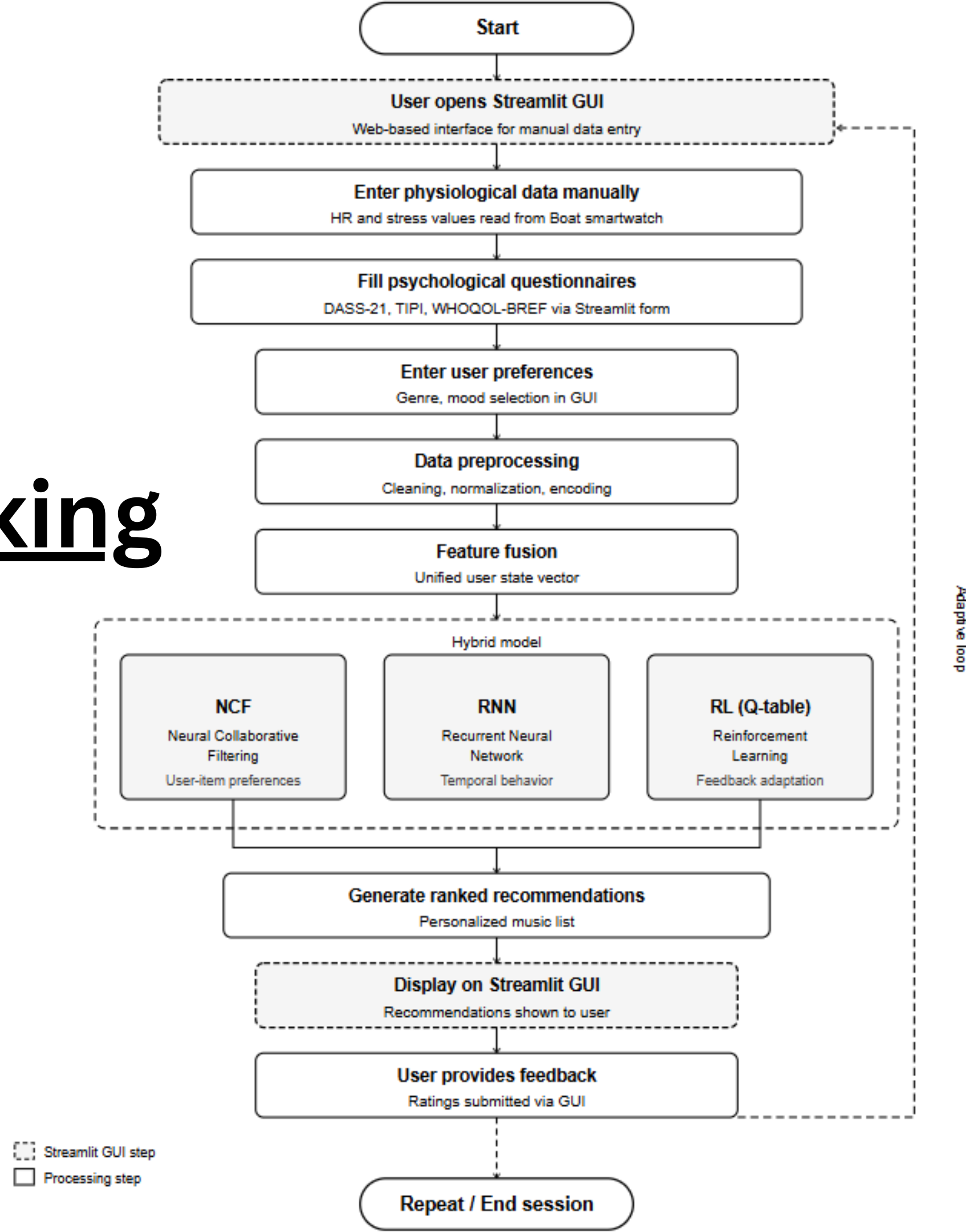
Application Domains

- Digital Therapeutics
- Smart Healthcare Systems
- Mental Wellness Platforms
- Assistive Technologies

System Block diagram



System Working



Result and Analysis

 **Recommendations**

[Get Recommendations](#)

1. Isharon Isharon Men Dil Lenewale – Asha Bhosle;Mohammed Rafi

Rate this song (1 = Strongly dislike, 5 = Strongly like)

☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

[Submit Feedback for Song 1](#)

[Open Song 1 in Spotify](#)

2. Thandi Hawa Yeh Chandni Suhani – Kishore Kumar

Rate this song (1 = Strongly dislike, 5 = Strongly like)

☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

[Submit Feedback for Song 2](#)

[Open Song 2 in Spotify](#)

3. Aap Ki Nazron Ne Samjha – Lata Mangeshkar

Rate this song (1 = Strongly dislike, 5 = Strongly like)

☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

[Submit Feedback for Song 3](#)

[Open Song 3 in Spotify](#)

Final Top 5 music recommendations after analyzing user's emotional state and music preferences.

4. Tumko Hamari Umar Lag Jaye – Lata Mangeshkar

Rate this song (1 = Strongly dislike, 5 = Strongly like)

☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

[Submit Feedback for Song 4](#)

[Open Song 4 in Spotify](#)

5. Aa Ab Laut Chalen – Mukesh;Lata Mangeshkar

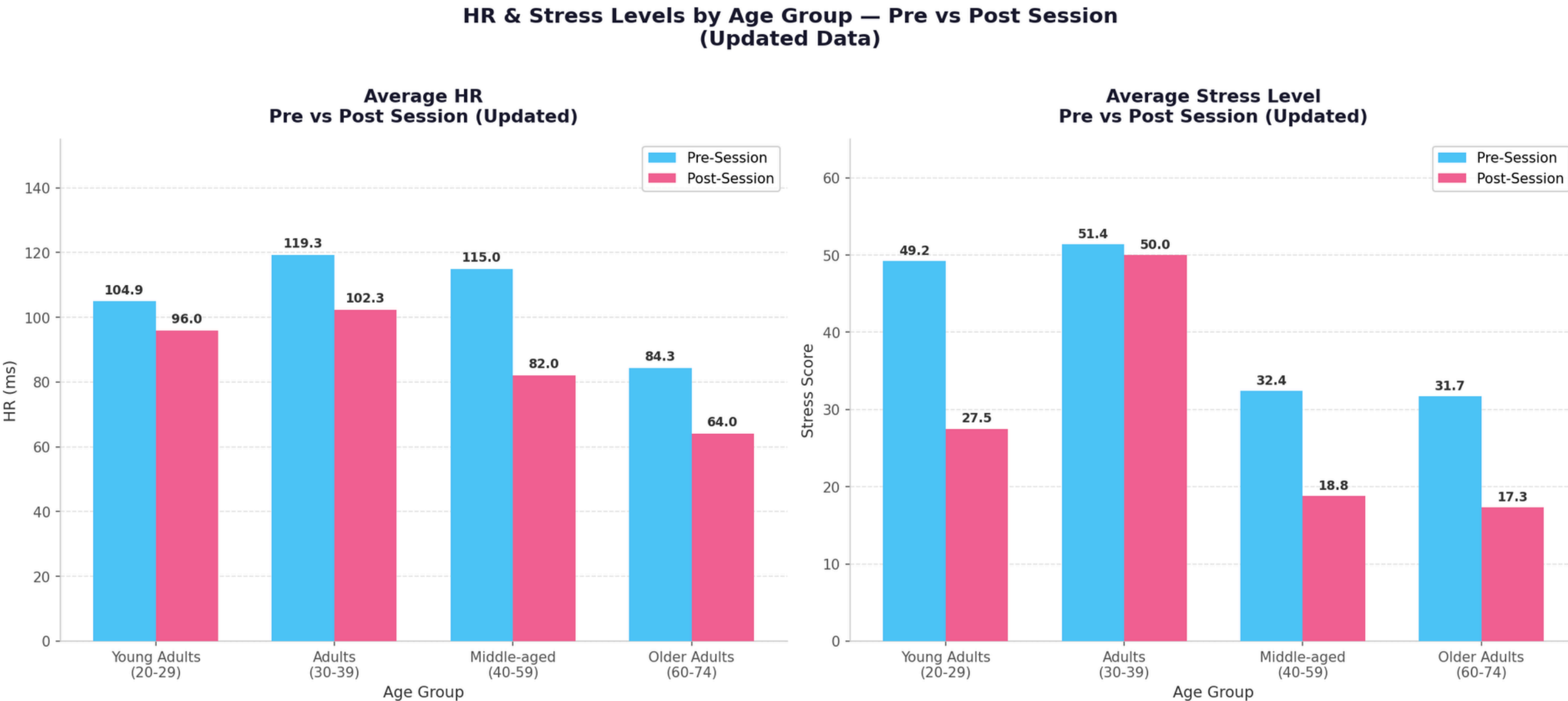
Rate this song (1 = Strongly dislike, 5 = Strongly like)

☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

[Submit Feedback for Song 5](#)

[Open Song 5 in Spotify](#)

The Average HR(Heart Rate) and Average Stress Level (Pre vs Post Session across Age Groups)



Age Group	HR Change	Stress Change	Interpretation
Young Adults	Moderate decrease	Strong decrease	Good responsiveness
Adults	Moderate decrease	Minimal change	Needs better personalization
Middle-aged	Large decrease	Strong decrease	Highly responsive
Older Adults	Large decrease	Strong decrease	Most benefited

Formula:

$$\text{Stress Reduction (\%)} = \left(\frac{\text{Pre-session Stress} - \text{Post-session Stress}}{\text{Pre-session Stress}} \right) \times 100$$

Observed Results:

- Young Adults: ~ 44%
- Middle-aged Adults: ~ 42%
- Older Adults: ~ 45%

The standard classification metrics used to evaluate how well a machine learning model or diagnostic test is performing

The standard formulas for calculating the standard classification metrics.

- **Accuracy : 73.89%**
- **Precision : 73.80%**
- **Recall : 82.50%**
- **F1-Score : 84.68%**

Whereas,

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

1. Accuracy

The proportion of total predictions that were correct.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

2. Precision

The proportion of positive identifications that were actually correct.

$$\text{Precision} = \frac{TP}{TP + FP}$$

3. Recall (Sensitivity)

The proportion of actual positives that were identified correctly.

$$\text{Recall} = \frac{TP}{TP + FN}$$

4. F1-Score

The weighted average (harmonic mean) of Precision and Recall.

$$\text{F1-Score} = 2 \times \left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \right)$$

Alternatively, written in terms of the confusion matrix:

$$\text{F1-Score} = \frac{2TP}{2TP + FP + FN}$$

For the evaluation of the difference in system's performance with regard to various age groups, a one-way **ANOVA** test was run on mean values of stress reduction among young, middle-aged and elderly test subjects. The output obtained was quite close to unity, with a high within group variance, as compared to between- group variance, **and p less than 0.05**.

Use cases and Costing

Use Case

Use Case Areas	Applications of the Music Recommendation System
Mental Wellness & Stress Relief	The system can be used by individuals experiencing stress or anxiety to receive calming and mood-regulating music recommendations based on emotional analysis.
Educational & Work Environments	Students and professionals can use the system to improve focus, concentration, and productivity through adaptive study/work playlists.
Healthcare & Therapy Support	The system can assist psychologists and wellness centers by providing emotion-aware music therapy support using questionnaire-based emotional assessment.
Daily Personalized Entertainment	Users can receive real-time personalized music recommendations according to their mood, personality traits, and emotional well-being.

Cost

Component	Model	Cost (INR)
Smartwatch	boAt Lunar Seek B359	1499
Headphones	HP B4B09PA Wired Headphones	749
Total Cost		2248

Role of every member

Adithyakrishna S	AI architecture design, data preprocessing, model optimization, and adaptive recommendation implementation,Refence paper editing
Keziah Vinod	AI model development, Reinforcement Learning integration, dataset generation,training/testing, and report editing
Dnyana Navale	Research paper editing, reference model preparation (using CGM input), and data collection for training model
Aadesh labde	Documentation, workflow design, data collection and implementation coordination

Conclusion and Future Scope

- **Conclusion**

The Adaptive Learning Multimodal AI Music Recommendation System successfully provides personalized and emotion-aware music recommendations using physiological and psychological data. By integrating machine learning with real-time mood detection, the system improves user experience, emotional well-being, and recommendation accuracy in a scalable and low-cost manner

- **Future Scopes**

1. **Integration of Advanced Physiological Sensors:** Future enhancements may include the incorporation of EEG, GSR, and real-time HRV monitoring to enable more accurate and continuous emotional state detection, thereby improving mood classification and reliability.
2. **Cloud-Based and Cross-Platform Deployment:** The system can be extended into a scalable cloud-based architecture with API integration, enabling seamless access through mobile applications.
3. **Advanced Reinforcement Learning Techniques:** Implementation of advanced reinforcement learning approaches such as Deep Q-Networks (DQN)
4. **Validation at hospital for patients with chronic stress related issues.**

Reference Paper/Project Competetion

Paper title: Adaptive Learning Multimodal AI Music Recommendation System

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Drafted in IEEE format and yet to publish after hospital validation.

Got Runner Up(Software Domain) in Tantragyan-National level Project Competition held by Lokmanya Tilak college of engineering,Khoparkhairne under software section



References

- [1] A. Mate, P. Narkhede, P. Kulkarni, H. Lokhande, J. Prasad, and R. Prasad, “Musical therapy for stress reduction using machine learning and IoT,” in Proc. 2024 MIT Art, Design and Technology School of Computing International Conference ([MITADTSOciCon](#)), Pune, India, 2024. pp. 1-6,
- [2] J. [Torous](#), R. [Jän Myrick](#), A. Rauseo-Ricupero, and K. H. Firth, “Digital mental health and COVID-19: Using technology today to accelerate the curve on access and quality tomorrow,” JMIR Ment. Health, vol. 7, no. 3, p. e18848, 2020.
- [3] Y. P. Pingle, “Machine Learning Techniques in Music Recommendation System for Healthcare Management,” Int. J. of Res. and Dev. in Comp. Sci. and Tech., vol. 8, no. 1, pp. 1-8, Jan. 2024. (Assuming a journal/periodical format for a review paper).
- [4] H. Guo et al., “EMO-Music: Emotion [recognition](#) [based](#) music therapy with deep learning on physiological signals,” in Proc. 2024 IEEE 1st Int. Conf. [Artif. Intell. Med.](#), Health and Care (AIMHC), Laguna Hills, CA, USA, 2024, pp. 10–13, [doi: 10.1109/AIMHC59811.2024.00008](#).
- [5] A. Karve, A. [Humnabadkar](#), B. [Shivbhakta](#), and S. Naik, “Machine learning based mood prediction and recommendation system,” Int. Res. J. Eng. Technol. (IRJET), vol. 8, no. 9, pp. 1–10, 2024.

[6] Y. P. Pingle, “Machine Learning Techniques in Music Recommendation System for Healthcare Management,” *Int. J. of Res. and Dev. in Comp. Sci. and Tech.*, vol. 8, no. 1, pp. 1-8, Jan. 2024. (Assuming a journal/periodical format for a review paper).

[7] D. Chanda and D. Levitin, “The neurochemistry of music,” *Trends Cogn. Sci.*, vol. 17, no. 4, pp. 179–193, 2013.

[8] S. K. Shravanthi, S. R. Sneha, S. Joshi, and V. R. Vandana, “A personalized music recommendation system,” *Int. Res. J. Eng. Technol. (IRJET)*, vol. 6, no. 6, pp. 1–4, 2019.

[9] H. Guo et al., “EMO-Music: Emotion recognition based music therapy with deep learning on physiological signals,” in *Proc. 2024 IEEE 1st Int. Conf. Artif. Intell. Med., Health and Care (AIMHC)*, Laguna Hills, CA, USA, 2024, pp. 10–13, doi: 10.1109/AIMHC59811.2024.000008.

[10] A. Karve, A. Humnabadkar, B. Shivbhakta, and S. Naik, “Machine learning based mood prediction and recommendation system,” *Int. Res. J. Eng. Technol. (IRJET)*, vol. 8, no. 9, pp. 1–10, 2024.

THANK



YOU!